

# Product Requirements Document (PRD)

## ScorePlus — Pratham Test Series & Center-Based CBT

**Document Status:** Final Consolidated Functional PRD

**Version:** 3.0 — Consolidated Final (supersedes Version 2.0)

**Product:** ScorePlus Pratham / Test Series + Center-Based CBT

**Platforms:** Existing ScorePlus Laravel Admin Panel, ScorePlus Website, ScorePlus Mobile App, existing Node.js Windows CBT Desktop Application

**Typography:** Poppins

**Database Strategy:** Existing MySQL remains for existing Form Builder configuration. After a CBT-enabled form is submitted, Pratham/Test Series operational data for **both Online and Center-Based packages** uses PostgreSQL.

## 0. What Changed from Version 2.0

Version 3.0 intentionally replaces the old Home-vs-Center desktop architecture.

| Area                      | Version 2.0                                               | Version 3.0                                                                                                       |
|---------------------------|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Exam modes                | Center Mode + Home Mode, both through the Windows CBT app | <b>Online</b> (Website/Mobile App) or <b>Center-Based</b> (existing Node.js Windows CBT app at a physical center) |
| Home Mode via desktop app | Supported                                                 | <b>Removed. Must not survive in implementation.</b>                                                               |
| Product naming            | CBT / Test Series                                         | <b>Pratham = Test Series</b> — one ecosystem                                                                      |
| Package type              | Not defined                                               | Exactly one Admin-selected type per package (dropdown): Online or Center-Based; no mixing                         |
| Mobile App                | Not in scope                                              | <b>Pratham</b> card on App Home, above Recent Videos / Top Videos                                                 |
| Duplicate detection       | Global, by identifying fields                             | Scoped to <b>candidate + same package</b>                                                                         |
| Form fill-up              | Per registration                                          | <b>Once per package</b>                                                                                           |
| Payment                   | Package purchase implied                                  | Explicit step: Form → Duplicate Check → City Preferences → Payment → Entitlement                                  |
| City / center             | Center assigned per candidate                             | Package-level city preference; physical center may differ per test; capacity, groups, slots                       |

| Area                             | Version 2.0                             | Version 3.0                                                                                                                                   |
|----------------------------------|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Scheduling conflicts             | Not defined                             | Cross-package clash checks at allocation and after every manual reschedule                                                                    |
| Allocation                       | Direct assignment                       | Tentative → Admin <b>Finalize Allocation</b> → per-exam admit card → date/slot/center locked                                                  |
| Identity verification            | Face-compare (Home) / physical (Center) | <b>Fresh invigilator verification for every Center-Based live exam</b> , using only data the form collected                                   |
| Emergency pause                  | Policy to be defined                    | End time extended by the <b>exact authorized pause duration</b>                                                                               |
| Result release                   | New per-exam release scheduler          | <b>Reuse existing release flow</b> , plus a gate: all assigned slots of that exam closed                                                      |
| Practice                         | Same CBT-style interface                | Always on <b>Website/Mobile App</b> , never on the desktop app                                                                                |
| Late purchase                    | Not defined                             | Allowed; finished exams become Practice where eligible; upcoming exams stay Live                                                              |
| Language                         | Not defined                             | Fixed per test by Admin; separate test entries per language                                                                                   |
| Camera/face/gaze proctoring      | Home + Center (configurable)            | Never for Online; only optional/configurable for Center-Based                                                                                 |
| Absent candidates                | Status only                             | Official status; never blocks slot closure or result release                                                                                  |
| Package cancellation/refund      | Not defined                             | <b>None in this scope</b> (different from individual attempt invalidation)                                                                    |
| Desktop app                      | "Windows CBT app"                       | The <b>existing</b> Node.js Windows CBT app is extended. No new desktop app.                                                                  |
| Question leakage across slots    | Not defined                             | Admin picks per exam: Shuffle Same Questions / Random Subset from Larger Pool / Multiple Paper Sets; locked once the exam starts              |
| Center unusable after admit card | Not defined                             | Super Admin-only <b>Emergency Reallocation</b> with revoked + reissued admit card                                                             |
| Slot closure                     | Not defined                             | Auto-close when all candidates are terminal; Controller (assigned scope) / Admin (admin scope) / Super Admin (global) force-close with reason |
| Payment                          | Not defined                             | Payment state machine, idempotent webhooks, reconciliation for "debited but webhook pending"                                                  |
| Capacity                         | "Configurable"                          | Concurrency-safe allocation; capacity can never be exceeded                                                                                   |

# 1. Product Overview

**Pratham = Test Series.** They are one product ecosystem, not separate modules.

Students discover and purchase packages from:

- The existing **Website Test Series page**.
- A **Pratham** card on the **Mobile App Home page**, placed directly above Recent Videos / Top Videos.

Both surfaces use the same catalogue, pricing, schedules, entitlements and backend data.

The system extends the existing ScorePlus platform rather than replacing it. The existing Laravel Admin Panel remains the administrative interface, and a Pratham/CBT module is added to it.

Live exams run through one of two routes, decided by the package type:

- **Online Live Tests** — attempted on the ScorePlus Website or Mobile App.
- **Center-Based Live Exams** — attempted only at an assigned physical center through the **existing Node.js Windows CBT Desktop Application**.

All Practice Tests (for both package types) run on the Website/Mobile App.

The system is designed around these principles:

1. Pratham and Test Series are one ecosystem with one catalogue.
2. Reuse the existing ScorePlus Form Builder, test/question structure, marking/negative-marking logic, Live Test → Result → Practice behavior, admit-card delivery scheduling and result-release flow. Do not rebuild them.
3. Keep the Center-Based live examination isolated in the existing secure desktop application.
4. Give authorized center staff (Center Controller and Invigilator) operational control for Center-Based exams.
5. Run optional camera/proctoring intelligence locally where practical (Center-Based only) and send only required events/statuses to the backend.
6. Keep all Pratham/Test Series operational data, for both package types, in PostgreSQL after CBT-form submission. No competing sources of truth.

---

## 2. Goals

The system shall:

- Present Pratham/Test Series packages on the Website and in the Mobile App from one catalogue.
- Sell tests only as packages, through a cart-style flow that allows multiple packages.
- Let Admin set each package as Online or Center-Based through a dropdown.
- Allow an existing ScorePlus form to be marked as a **CBT Test Form**.
- Receive registrations from that form, once per package.
- Detect duplicate registrations scoped to candidate + package.
- Collect package-level city preferences for Center-Based packages.
- Take payment and grant package entitlement.
- Allocate Center-Based candidates to city, physical center, date, slot and group within capacity, without cross-package clashes.
- Require explicit Admin allocation finalization before admit cards.
- Generate a separate admit card per Center-Based live exam, and lock the assignment after generation.
- Generate roll numbers and passwords dynamically for Center-Based desktop login.

- Require fresh invigilator verification for every Center-Based live exam.
  - Conduct Center-Based live exams in the existing secure Windows CBT application.
  - Conduct Online live tests on the Website/Mobile App.
  - Randomize questions reproducibly.
  - Reuse the existing ScorePlus question/test model, including marking and negative-marking configuration.
  - Support NTA-style CBT navigation and question-palette behavior.
  - Provide optional local proctoring for Center-Based exams with controlled warning escalation.
  - Provide a unified Center Controller dashboard for Center-Based operations.
  - Recover safely from network, application and machine failures.
  - Handle late purchase and absent candidates without blocking others.
  - Calculate results automatically and release them through the existing release flow.
  - Generate a detailed candidate result/answer-analysis PDF.
  - Provide Practice Mode on Website/Mobile App.
  - Provide Live-only leaderboards.
  - Maintain complete audit logs.
  - Keep Pratham/Test Series lifecycle data in PostgreSQL.
- 

## 3. Explicitly Out of Scope

The following are **not part of this PRD**:

- **Home Mode through the Windows desktop app** (removed in v3.0).
- A new/second Node.js CBT desktop application.
- Package cancellation or refund workflow.
- Selling individual tests separately.
- Mixed packages (some tests Online, some Center-Based).
- Candidate-side language switching inside a test.
- Separate manually authored question papers for each time slot.
- A new result-release scheduler.
- A new admit-card delivery scheduler.
- New mandatory photo, Aadhaar, document or verification fields on the form.
- A system-level student-consent workflow for moving to Second Preference city.
- Separate CVT Manager role.
- AI Question Generator.
- PYQ-specific mock-test workflow as a separate CBT feature.
- Create Your Own Test feature.
- Separate subject-batch-exam mapping created specifically for CBT when equivalent configuration already exists.
- Free-test PDF export as a separate CBT feature.
- AdMob-related functionality.
- Separate Home and Center Exam Controller panels.
- Per-question time limits.
- A new independent negative-marking or scoring system.
- Unnecessary duplication of existing ScorePlus test-builder functionality.

- A separate student account system.

Existing ScorePlus capabilities may still be reused when required; they are simply not being rebuilt as separate modules.

---

## 4. System Components

### 4.1 Existing ScorePlus Laravel Admin Panel

The existing Laravel Admin Panel remains the central administrative product. A **Pratham/CBT section/module** is added for:

- Package creation and management (name, description, price, type dropdown, included tests, subjects/modules, schedules, language, applicable cities, availability).
- Test creation and management (reusing the existing test/question structure).
- Candidate registrations and duplicate-registration review.
- Payments/enrollment and entitlements.
- City preferences, cities, physical centers and capacity.
- Groups, dates, time slots and allocation (tentative → finalized).
- Cross-package conflict review.
- Candidate credentials (Center-Based).
- Admit-card configuration and generation.
- Center Controller and Invigilator account management and assignment.
- Live examination monitoring (Center-Based).
- Proctoring configuration (Center-Based).
- Emergency pause oversight.
- Results, result-release configuration (existing flow) and answer-analysis delivery.
- Practice eligibility.
- Leaderboards.
- Audit logs.
- Reports and exam summaries.

All Pratham/CBT operational data must read from/write to PostgreSQL.

### 4.2 ScorePlus Website

- Existing **Test Series page** shows Pratham packages as product-style cards.
- Cart-style purchase of one or more packages.
- CBT-enabled form fill-up, city preference (Center-Based), payment.
- Existing Student Dashboard receives the Pratham/Test area.
- Online Live Test attempts.
- Practice Test attempts (both package types).
- Results, answer analysis and leaderboards.

- For Center-Based packages: registration, preference, payment, allocation, admit-card and status views only — **never** the Center-Based Live Exam attempt interface.

## 4.3 ScorePlus Mobile App — Pratham Entry

- A **Pratham** card on the App Home page directly above Recent Videos / Top Videos.
- Uses the same Test Series catalogue as the Website.
- Students can browse, view details, purchase, see Online/Center-Based status, view language/schedule/city details, access entitled Online Live Tests and access Practice Tests.
- For Center-Based packages, the app may handle/view registration, preference, payment, allocation, admit-card and status information, but must **never** provide the Center-Based Live Exam attempt interface.

## 4.4 Existing Node.js Windows CBT Desktop Application

The existing desktop app is **dedicated to Center-Based Live Exams**. No new desktop application is created; the existing one is modified/extended.

Responsibilities:

- Candidate login (Roll Number + Password).
- Pre-exam waiting/countdown screen.
- Waiting for invigilator verification/authorization.
- Secure examination/kiosk mode.
- NTA-inspired CBT examination UI.
- Answer capture.
- Local temporary answer persistence.
- Server queue synchronization.
- Local proctoring detection where enabled.
- Warning display and flag generation.
- Crash/reconnect recovery.
- Session/device controls.
- Emergency pause handling.
- Auto-finalization and final examination state handling.
- Authorized center operations.

It is **not** used for Online Live Tests or Practice Tests, and it is **not** the question-management or administrative system.

## 4.5 Unified Center Controller Dashboard

One controller workflow manages all Center-Based candidates across assigned exams and centers. No separate controller products are required. Online Live Tests do not use Center Controller authorization or monitoring.

## 4.6 Invigilator Dashboard

Used at the physical center to verify candidates for each Center-Based live exam and record Present/Verified or Absent (see Section 20).

---

## 5. Package Model

### 5.1 Package-Only Commercial Model

The commercial model is **package-only**. Individual tests are not sold separately.

Example: **NEET Test Series 2026 — ₹8,000**

A package can contain any configured number of tests (e.g. 20, 30, 40, 50). Each included test remains an individual examination instance operationally, but the student purchases the entire package.

Admin configures package name, description, price, type, included tests, subjects/modules/exam grouping, individual exam schedules, language, applicable cities (Center-Based) and availability. Admin preconfigures Week 1, Week 2 and later tests.

Students can add multiple packages to a cart and purchase them together, including packages from different exam categories (e.g. NEET and JEE Main).

### 5.2 Package Type

Every package has exactly one Admin-selected type, chosen manually through a dropdown:

- **Online**
- **Center-Based**

There is no mixing. If a package is Center-Based, every included live test is Center-Based. If Online, every included live test is Online. The package type determines the execution route.

### 5.3 Package Cards and Details

Package cards/details must clearly show:

- Package name.
- Price.
- Number of tests.
- Subjects/modules.
- Available schedule.
- Test language.
- Online/Center-Based type.
- Applicable cities (Center-Based packages).

The UI must clearly state whether live tests are available on web/app or require attendance at a physical center.

## 5.4 Entitlement

The entitlement hierarchy is:

**Student Account → Purchased Package(s) → Included Test(s) → Scheduled Live Exam Instance(s)**

After purchase:

- All included tests map automatically to the student's account.
- No additional purchase is required for individual tests.
- Each scheduled Live Exam continues through its route (Online or Center-Based), result and leaderboard flow.

Package ownership controls access to the included exams. Package/test composition and entitlement must remain auditable.

## 5.5 No Cancellation / Refund

There is no package cancellation/refund workflow in this scope. Once purchased, the package remains purchased.

This is different from **individual exam-attempt invalidation/termination**, which an authorized center operator may still perform where approved exam rules require it (see Section 27).

---

# 6. Test Language

Language is fixed when Admin creates each test. A test may be English, Hindi, Bengali or another configured language.

No candidate-side language switch is required. Title, description and package information must make the language clear before purchase.

If the same exam is required in multiple languages, separate test entries are created.

---

# 7. CBT Form Builder Integration

The existing ScorePlus Form Builder must be **reused, not rebuilt**.

Add a checkbox to form configuration:

**[ ] Use this form as a CBT Test Form**

When unchecked:

- Existing form behavior remains unchanged.

When checked:

- Successful submission starts the PostgreSQL Pratham/CBT lifecycle.
- Submission is treated as a package registration.
- Scheduling/enrollment workflow becomes available.

Admin keeps full flexibility over form fields. **No photo, Aadhaar, document or verification-specific field is newly mandatory.** Invigilator verification later uses only the information actually collected by the form.

---

## 8. Account, Registration and Purchase Sequence

### 8.1 Final Sequence

**ScorePlus Account → CBT-enabled Form Fill-up → Submit → Package-Scoped Duplicate Check → City Preferences (Center-Based only) → Payment → Package Entitlement → Scheduling/Allocation**

### 8.2 Account

- A ScorePlus account is mandatory.
- Existing supported login methods continue: Email/Password, Phone/OTP, Google and Truecaller.
- No separate student account system is created.
- No separate Practice login page is created; Practice uses the student's ScorePlus account on the Website/Mobile App.

### 8.3 Registration Rules

- The registration form is filled **once per package**, not once per included exam.
- Incomplete/unpaid registrations must not enter final exam allocation and must not receive an admit card.

### 8.4 Payment Lifecycle and Reconciliation

Payment states:

`Initiated → Pending → Success → Failed`

plus the special state:

`Debited - Confirmation Pending` (gateway may have taken money, but ScorePlus has not yet received/verified the webhook or callback).

Rules:

- Entitlement is granted only on a **verified Success** (webhook or server-side status check with the gateway), never on the client redirect alone.

- Webhook/callback processing is **idempotent**: the same payment event received more than once grants entitlement only once.
- While a payment for the same cart/package is `Pending` or `Debited - Confirmation Pending`, the student must not be pushed into a second payment. The UI shows "Payment is being confirmed" instead of a new Pay button.
- A scheduled **reconciliation job** asks the gateway for the status of every Pending/Confirmation-Pending payment and moves it to Success or Failed.
- If a duplicate payment still happens, it is flagged for Admin review (refund handling is outside this system; package ownership is not affected).
- Admin can see payment state and gateway reference for each registration and can trigger a manual status re-check.
- All payment state changes are audited.

## 8.5 Step-by-Step Flow

1. Admin creates/configures a form using the existing Form Builder.
2. Admin checks **Use this form as a CBT Test Form** and links it to the package.
3. Student signs in with a ScorePlus account and selects package(s) on Website or App.
4. Student fills the form once for each package.
5. Student submits; the submission enters PostgreSQL.
6. Package-scoped duplicate detection runs.
7. Center-Based only: student gives city preferences.
8. Student pays (cart-style, one or more packages).
9. Package entitlement is granted; included tests map to the account.
10. Online: scheduled Online Live Tests appear in the Student Dashboard.
11. Center-Based: allocation (city/center/group/slot) → clash check → Admin adjustments → Finalize Allocation.
12. Center-Based: roll number and password generated; individual admit card generated per exam; assignment locked.
13. Admit card delivered through the existing delivery schedule.
14. Candidate attends the exam (Online on web/app; Center-Based at the center).

---

## 9. Duplicate Registration Handling

Duplicate detection is scoped to **candidate + same package**.

A student registering for NEET and for JEE is valid and must **not** be globally marked duplicate.

Within the same package, the system evaluates likely duplicates using configured identifying fields, for example:

- ScorePlus account.
- Email.
- Phone.
- Date of birth.
- Candidate identity fields.

- Other configured unique identifiers.

The system keeps only one valid active registration per candidate per package according to the configured policy.

Duplicate submissions are marked/cancelled with a reason such as:

#### **Cancelled - Duplicate Registration**

Admin must be able to review duplicate decisions.

A cancelled duplicate must not receive a second entitlement, roll number, allocation, exam attempt or admit card.

---

## 10. Late Purchase

There is no cut-off just because some package tests have already ended.

If a student purchases late:

- Finished live exams are not recreated for that student.
  - Eligible finished exams become available as Practice according to the existing publication/practice flow.
  - Upcoming exams continue normally as Live Exams.
  - The student gets no retrospective official rank for a Live Exam they did not attend.
- 

## 11. Online Package

Online Live Tests are attempted through the existing ScorePlus Website or Mobile App.

The existing Student Dashboard receives the Pratham/Test area; no separate student account system is created.

Online tests do **not** use:

- Physical invigilator verification.
- Windows desktop CBT app.
- Kiosk mode.
- Camera/face/gaze proctoring.
- Center Controller authorization.
- Physical center allocation.
- City preference.

Existing Live Test → Result → Practice behavior is reused where applicable. Online operational data still lives in PostgreSQL.

---

## 12. Center-Based Package

Center-Based packages may be viewed and purchased from both Website and App, but their **Live Exams can only be attempted at an assigned physical center using the existing Node.js Windows CBT Desktop Application.**

No new desktop application is created; the existing application is modified/extended.

After a Center-Based live exam is completed and becomes eligible for Practice, its Practice version is attempted on Website/Mobile App, never on the center desktop app.

---

## 13. City Preference (Center-Based)

- City preference is collected **once at package level**.
- Students can give First Preference, Second Preference and other configured preferences.
- The allocated city applies across the whole package. Students do not change city for individual included tests.
- **First Preference has maximum priority.** If capacity is full, the system/Admin should first keep the same city by using additional/later slots or dates. **There is no maximum delay limit.**
- **Second Preference is a rare fallback.** Admin may manually move selected students/groups to Second Preference without an additional system-level student-consent workflow.
- When a student/group is moved to Second Preference (or any other city), the new city applies to **all remaining Center-Based exams of that package** whose admit cards are not yet generated. Exams whose admit cards are already generated stay locked (Section 17.4) unless changed through Emergency Reallocation (Section 17.6).
- Admin may manually tweak dates for holidays, festivals, capacity or operational needs.
- All overrides are audited.

---

## 14. Physical Centers and Capacity

- The city is package-level, but the exact physical center may differ between included exams. Example: Test 1 at Udaipur Center A, Test 2 at Udaipur Center C.
- Each center has configurable seat/computer capacity.
- **Capacity must never be exceeded, even under concurrent actions.** Allocation to a center/date/slot must be atomic (database-level locking, conditional update or an equivalent concurrency-safe method). Example: capacity 20, 19 allocated, two Admins each add one student at the same moment → exactly one succeeds, the other gets a "slot full" error. The same applies to bulk/auto allocation, reschedules and Emergency Reallocation.
- The system supports multiple centers, dates, time slots and candidate groups so hundreds or thousands of candidates can be spread safely.
- Admin sees total, allocated, pending, completed, absent and unallocated students, plus capacity by center/date/slot.

Admit cards can show center name, address/location and reporting information.

---

## 15. Cross-Package Scheduling Conflicts

- A student may own multiple Center-Based packages.
  - The system must prevent overlapping exams for the same student.
  - A 1–2 day gap between a student's Center-Based exams is preferred where feasible but is **not mandatory**.
  - Conflict checking runs during allocation **and again after every manual Admin reschedule**.
  - A conflicting manual change must show a warning and must be resolved before allocation finalization/admit-card generation.
- 

## 16. Exam Scheduling and Configuration

Every included test/exam has its own independent configuration:

- Exam name.
- Package and package type (inherited).
- Language.
- Exam date(s).
- Time slot(s) with scheduled start/end time/duration.
- Candidate assignment/groups (Center-Based).
- Center(s) and capacity (Center-Based).
- Question/test configuration and question pool/pattern.
- Question delivery strategy (Section 23.2).
- Result-release configuration (existing flow).
- Practice eligibility.
- Proctoring configuration (Center-Based).
- Controller/Invigilator assignment (Center-Based).

Multiple exams may run at the same time. Settings must be exam-specific, not one global settings page for all exams.

Each included test has an **independent lifecycle**.

---

## 17. Tentative Allocation, Finalization and Admit-Card Lock

### 17.1 Tentative Allocation

Before admit-card generation, allocations are tentative. Admin may change date, slot, physical center, group and permitted city fallback as many times as needed. Every change reruns the clash check (Section 15).

### 17.2 Finalize Allocation

Admin must explicitly **Finalize Allocation** before admit-card generation/scheduling. Finalization is blocked while unresolved conflicts exist or while any included registration is unpaid or incomplete.

## 17.3 Per-Exam Admit Card

Each individual Center-Based live exam receives **its own admit card** as a professionally formatted PDF.

The admit card may contain:

- Candidate name and information.
- Roll number.
- Package name.
- Test/exam name.
- Language.
- Exam date.
- Reporting/start time.
- City.
- Physical center name and address.
- Candidate instructions.
- Relevant submitted form data.
- Other configured candidate/exam fields.

## 17.4 Assignment Lock

**Once an individual exam's admit card is generated, its date, slot and physical center are locked and cannot be changed** through normal editing. The only exception is Super Admin Emergency Reallocation (Section 17.6).

The lock is **per exam, not package-wide**. Example: if Test 1's admit card is generated while Tests 2 and 3 are still tentative, Test 1 is locked and Tests 2 and 3 stay editable.

## 17.5 Delivery

Existing admit-card delivery scheduling is reused (e.g. immediately after eligibility, one day before, two days before, one week before, or a custom time, as the existing system supports). The admit card is emailed and also visible on Website/App. Delivery status must be recorded.

## 17.6 Emergency Reallocation (Super Admin Only)

For genuine emergencies after admit-card generation — center closed, fire, flood, power/infrastructure failure, computer lab unavailable, natural disaster or administrative closure — a **Super Admin-only exceptional action** called **Emergency Reallocation** is available.

It is not a normal edit. Normal Admins cannot use it.

Requirements:

- Only Super Admin can perform it.
- Mandatory reason (and incident category).
- Scope can be a single candidate, a group, a slot or a whole center for that exam.

- The new date/slot/center must respect capacity (Section 14) and pass the cross-package clash check (Section 15). The allocated city rules (Section 13) still apply.
- The previous admit card is **revoked** (marked invalid, no longer usable for entry or desktop login binding).
- A new assignment version and a new admit-card version are created; old versions stay in history.
- Credentials are rebound to the new assignment where needed.
- The student is **notified** (email and Website/App) with the new admit card.
- Invigilators/Controllers of both old and new centers see the change.
- Complete audit history is preserved: actor, reason, old and new assignment, old and new admit-card versions, timestamps.

Emergency Reallocation cannot be used after a candidate has started the exam; for in-exam failures use Emergency Pause (Section 24.4).

---

## 18. Candidate Credentials (Center-Based)

For each valid Center-Based candidate:

- Roll number is generated dynamically.
- Password/credential is generated dynamically.
- Credentials are bound to the assigned examination.
- Credential lifecycle must be auditable.
- Passwords must not be stored as readable plaintext in the database.

Desktop login uses:

**Roll Number + Password**

Online tests and Practice use the student's ScorePlus account on Website/App.

---

## 19. Roles and Permissions

### 19.1 ScorePlus Main Admin / Super Admin

Global administrative access. **Emergency Reallocation after admit-card generation (Section 17.6) is strictly reserved for Super Admin.** It is never available to Admin, Center Controller or Invigilator.

Slot force-close (Section 34.1) follows scope:

- **Admin:** can force-close any slot within their authorized administrative scope.
- **Super Admin:** can force-close any slot globally.
- Both require a mandatory reason and full audit logging; remaining non-terminal candidates are finalized per Section 34.1.

- Create/manage packages and set package type.
- Create/manage tests, questions and language.
- Manage registrations, duplicates, payments and entitlements.
- Configure cities, centers, capacity, dates, slots and groups.
- Allocate, reschedule, move to Second Preference, resolve conflicts and Finalize Allocation.
- Generate/schedule admit cards.
- Create Center Controllers and Invigilators and assign them.
- Configure proctoring (Center-Based).
- View all candidates and all controller/invigilator actions.
- Access all audit logs.
- Configure result release (existing flow) and practice eligibility.
- Review/invalidate individual attempts.
- Access reports.

## 19.2 Center Controller (Exam Controller)

Logs in with an ID/password issued through the administration system. Assigned specific Center-Based exams, centers and/or candidates.

For the assigned scope, the Center Controller has full operational authority:

- View candidates.
- See who has not logged in, is waiting, is awaiting verification, has started, is in progress, has submitted, is flagged, is terminated/cancelled or is absent.
- Allow a candidate to begin where manual authorization is required.
- Approve a reconnect/resume or device change.
- Review proctoring flags, clear a flag, allow the candidate to continue.
- Terminate/invalidate an exam attempt.
- Invalidate an attempt even after submission if fraud is later established.
- Enable/disable permitted proctoring controls where policy allows.
- Execute emergency center-level pause/resume.
- Force-close **only** slots within their assigned exam/center/scope, with a mandatory reason and full audit logging; remaining non-terminal candidates are finalized per Section 34.1.

The Center Controller has **no** Emergency Reallocation authority.

Every critical action requires a reason where appropriate and is audit logged with Controller ID and timestamp.

## 19.3 Invigilator

- Sees the assigned candidates for each Center-Based live exam.
- Views the candidate's submitted Form Builder data relevant to verification.
- Records **Present/Verified** or **Absent**.
- Has **no** slot force-close or Emergency Reallocation authority.

## 19.4 Student / Candidate

- Register with a ScorePlus account and fill the form per package.
- Give city preferences (Center-Based).
- Pay and receive entitlement.
- Receive credentials/admit card (Center-Based).
- Attempt Online Live Tests on Website/App.
- Attend the center and attempt Center-Based Live Exams on the desktop app after verification.
- Navigate questions, mark for review, clear responses, submit where the exam flow permits.
- Attempt Practice Tests on Website/App.
- Receive result/answer analysis when released.
- View leaderboards.

---

# 20. Invigilator Verification (Center-Based)

Fresh invigilator verification is required for **every individual Center-Based live exam**.

The Invigilator dashboard shows the candidate's submitted Form Builder data relevant to verification:

- If the form has photo/document fields, show them.
- If it only has basic information, use only that information.
- No additional mandatory verification fields are introduced.

The invigilator records **Present/Verified** or **Absent**.

Required verification must be complete before the candidate can start that Center-Based live exam.

If a candidate cannot be verified, the case goes to **Manual Verification Required** for the Center Controller, who may approve or reject. Decision, reason, actor ID and timestamp are audit logged.

Automated face matching is not required. If optional camera checks are enabled for a Center-Based exam, they only support the invigilator and are never the sole irreversible authority.

---

# 21. Absent Candidates

- **Absent** is the official Live Exam status.

- Absence does not block slot closure or result publication for other candidates.
  - An absent student has no official Live attempt, score or personal rank for that exam.
  - The published leaderboard may still be viewed.
  - The exam may later be available to that student as Practice under the normal practice rules.
  - **Once marked Absent for the assigned live slot, the candidate cannot be reassigned to another Live slot of the same exam** (same day or later). Only Practice eligibility remains.
  - Absent does not affect the student's other exams in the package.
- 

## 22. Test/Question Management

Reuse the existing ScorePlus test/question structure rather than rebuilding authoring logic. The PostgreSQL-backed Pratham/CBT management flow supports both package types.

Admin manually selects **Online** or **Center-Based** at package level using the dropdown; that type decides the execution route.

Supported question content follows existing ScorePlus capabilities:

- Text-only question.
- Question text + image.
- Image-based question.
- Text options.
- Image options.
- Text + image options where the existing structure supports it.
- Correct-answer configuration.
- Existing marking logic.
- Existing negative-marking logic.

Do **not** create a duplicate scoring/marking engine.

---

## 23. Randomization and Paper Versioning

### 23.1 Randomization

- Separate manually authored papers are **not** required for each time slot (unless Admin chooses Multiple Paper Sets, Section 23.2).
- Slots may use the same configured question pool/pattern.
- Candidate question delivery is randomized according to the approved existing model and the exam's selected delivery strategy.
- Randomization should be decided by backend/API logic rather than trusting only the client.
- The exact randomized assignment must be reproducible/auditable for later result verification.

- The desktop app (Center-Based) or Website/App (Online) receives the candidate-specific exam payload and renders it.

## 23.2 Question Delivery Strategy (Leakage Protection)

Random order alone does not stop leakage when one exam runs over many slots or days (e.g. 3,000 candidates over 2–3 days). Admin therefore selects one strategy **per exam**:

1. **Shuffle Same Questions** — every candidate gets the same questions in a randomized order (and option order where supported). Suitable for single-slot or small exams.
2. **Random Subset from Larger Pool** — each candidate gets a randomized subset from a larger question bank that follows the same pattern (subject/section counts, marks, difficulty mix as configured).
3. **Multiple Paper Sets** — Admin prepares multiple equivalent paper sets; sets are assigned per slot/day (or per candidate) as configured.

Rules:

- The selected strategy and its configuration are **locked once the exam's first slot starts**.
- Whatever the strategy, each candidate's exact paper (questions, order, set, version) is stored and reproducible for result verification and disputes.
- Scoring always uses the candidate's own assigned paper.
- Online Live Tests may use the same strategies.

## 23.3 Paper Versioning and Locking

Once an exam is published/locked for delivery:

- The delivered question-paper version must be identifiable.
- Silent edits must not alter an already assigned candidate attempt.
- Any authorized post-publish change must create a traceable version/change record.
- The candidate's assigned randomized paper must be reproducible for dispute/result review.

---

# 24. Timing Rules

- Server time is authoritative, not the candidate's editable local clock.
- There is **no per-question timer**; the exam has one overall official time window.
- Normal late start does not silently grant extra time.
- Switching questions, reconnecting, restarting the app or Controller authorization does not automatically give extra time.

## 24.1 Pre-Exam Login and Countdown (Center-Based)

Before the scheduled start:

- Candidate may open the CBT application and log in early.

- Application is not yet locked into secure mode; candidate may exit before activation.
- Questions must not be exposed.
- A countdown shows remaining time. Example: **Your examination starts in 00:18:42**

Online Live Tests likewise show a waiting/countdown state with no question exposure until the official start.

## 24.2 Exam Start Logic (Center-Based)

At the scheduled start time:

1. Candidate's valid session becomes eligible.
2. Fresh invigilator verification must already be recorded.
3. Any required proctoring checks run.
4. Exam payload becomes available.
5. Secure Exam/Kiosk Mode activates.
6. Candidate sees the CBT question interface.
7. Official timer runs against server time.

A candidate cannot start before the configured start/authorization conditions are met.

## 24.3 Controller-Authorized Start

In exceptional cases a candidate may need Center Controller authorization. The UI shows a message like:

***Please ask your Exam Controller to allow your examination.***

Authorization does **not** grant extra duration. The official end time stays unchanged unless an authorized Emergency Pause changes it.

## 24.4 Emergency Pause and Resume

For an authorized center-wide Emergency Pause caused by power, network or infrastructure failure (or another authorized operational emergency):

- Explicit confirmation is required.
- Mandatory reason.
- Affected center/exam/candidate scope.
- Server-authoritative pause state; candidate state is preserved.
- Controlled resume.
- The official end time is extended by the **exact authorized pause duration**.
- Reason, operator, start and resume timestamps are audited.

Emergency pause is different from an ordinary late start. Time must never be changed by the client.

---

## 25. Secure Exam / Kiosk Mode (Center-Based Only)

This is a **top-priority security requirement** for Center-Based live exams. It never applies to Online Live Tests or Practice.

Before exam start the application behaves normally.

Once the exam officially starts:

- Application enters secure edge-to-edge/full-screen mode.
- Candidate must not be able to casually leave the exam UI.
- Window/app switching attempts are blocked or detected where technically supported.
- Restricted shortcut/activity attempts create security events according to policy.
- Exam UI stays dominant until the exam is completed, terminated or officially released.

Controls to evaluate:

- Full-screen kiosk shell.
- Alt+Tab handling.
- Windows key handling.
- Task switching.
- Unauthorized application switching.
- Screen-capture/recording policy where technically feasible and legally appropriate.
- Multiple-instance prevention.

Controls must not make a machine unrecoverable if the app fails. Controller/admin recovery paths are mandatory.

---

## 26. NTA-Inspired Candidate Examination UI

The Center-Based desktop UI is **inspired by the familiar NTA CBT interaction model**, not an exact visual copy. Online/Practice on Website/App reuse the existing ScorePlus test UI.

Use **Poppins** throughout.

### 26.1 Core Layout

- Top header with exam/candidate information.
- Remaining-time display.
- Main question content area.
- Question options.

- Right-side question palette.
- Bottom navigation/actions.

## 26.2 Candidate Actions

- Save & Next.
- Clear Response.
- Mark for Review & Next.
- Direct question navigation through the palette where permitted.
- Final confirmation/summary where applicable.

## 26.3 Question Palette

The palette must visually distinguish:

- Not Visited.
- Not Answered.
- Answered.
- Marked for Review.
- Answered & Marked for Review.

Use clearly different **shape + color + label/legend**, so status is not shown by color alone. NTA-style badge ideas (grey unvisited, red not answered, green answered, purple review) may be used while keeping ScorePlus branding and accessibility. Each indicator shows the question number.

---

# 27. Attempt Persistence, Submission and Recovery (Center-Based)

## 27.1 Answer Persistence and Server Queue

Answers must not depend on a single final-submit request.

1. Response changes are saved to encrypted/secured local temporary state.
2. Changes sync to a server-side queue/state when connectivity exists.
3. Sync continues throughout the attempt.
4. Backend tracks candidate progress/state.
5. At the official finalization point, queued/current data is consolidated into durable result-processing state.

**The candidate machine must never be the only copy of exam progress.** A PC shutdown right after answering must not destroy the recorded attempt.

## 27.2 Submission and Exam End

At the official end:

- Remaining active attempts are auto-finalized.
- Queue/state is consolidated.
- Scoring begins.
- Candidate session is closed.
- Kiosk mode is released safely.

If a candidate has left or shut down the machine, synced server state must still be processable.

## 27.3 Crash, Restart and Reconnect

If the app crashes or the machine restarts:

- Candidate reopens the app and re-authenticates.
- System finds the existing active attempt.
- Local and server state are reconciled through the authorized reconnect flow.
- Candidate resumes from the last safely recorded state.
- Official end time is unchanged (no extra time for a crash).

## 27.4 Network Failure

- Candidate may continue where the exam's offline-resilience policy allows.
- Responses keep saving locally; sync retries automatically.
- On reconnect, pending data syncs; conflicts reconcile deterministically.
- Valid responses must not be lost due to temporary loss.
- UI shows connectivity state without distracting the candidate.

## 27.5 Device and Session Binding

- Active attempt is bound to an exam session/device identity.
- Concurrent candidate sessions are prevented.
- Login from another machine during an active attempt is blocked or needs Controller authorization.
- Controller-approved device change is audit logged.
- Previous session invalidation is controlled.

**Center machine replacement during an exam** (e.g. PC fails, candidate moved to another PC) must follow:

Old device session invalidated → new device authorized by Controller → answers restored  
from server state → original end time preserved → full audit

## 27.6 Attempt Invalidation (Not Package Cancellation)

Submission does not remove the Controller's fraud-review authority. If credible fraud/violation is found during or after the exam, an authorized Controller may terminate/invalidate that **individual attempt**.

Requirements: mandatory reason, actor identity, timestamp, previous status, new status, audit trail. Result/release and leaderboard must respect invalidated status.

This never cancels the student's **package ownership**.

---

## 28. Local Proctoring (Center-Based, Optional)

Camera/face/gaze proctoring is **never** used for Online Live Tests or Practice. For Center-Based exams it is optional and configurable.

Where enabled, detection runs locally in the desktop app as far as practical. Possible signals:

- Face absent for a sustained threshold.
- Multiple faces.
- Repeated abnormal head direction.
- Sustained/repeated gaze away from screen.
- Candidate leaving the seat/frame.
- Possible phone/object detection where supported.
- Repeated attempts to switch away from the exam app.
- Restricted-key/app-switch activity (applies with kiosk even without camera).
- Camera disabled/unavailable where camera is mandatory.

A single natural head movement or brief glance must not be treated as fraud. Use thresholds, duration and repeat logic to reduce false positives.

Raw continuous video is not sent to the backend. The backend receives required status/events and evidence metadata by policy.

### 28.1 Warning and Flag Escalation

Default for qualifying violations:

- **First:** Warning 1
- **Second:** Warning 2
- **Third:** Candidate becomes Flagged; Controller is alerted.

The Controller can review, allow to continue, clear the flag, keep monitoring, or terminate/invalidate the attempt.

Escalation can repeat by exam rule. Warnings are clear but not disruptive.

### 28.2 Proctoring Configuration

- Proctoring enabled/disabled per exam.
- Camera detection enabled/disabled.

- Candidate-specific exception where permitted.
- Center-specific configuration where needed.

If camera/proctoring is off, the candidate is not prompted for a camera. Kiosk security stays independent. Runtime security-setting changes are audit logged.

---

## 29. Pre-Exam System Check and App Version Enforcement (Center-Based)

### 29.1 Center Machine System Check

Before exam day, each center machine can be checked for:

- Supported Windows/app version.
- Camera availability/permission where required.
- Network and server connectivity.
- Display/resolution suitability.
- Required local services.
- Application integrity/version.

Failures must be actionable before the exam starts.

### 29.2 App Version Enforcement

The backend defines a minimum supported desktop version. If an outdated or incompatible version is found:

- The center/candidate is told before the exam.
  - Exam start can be blocked when unsafe.
  - Controller can see the failure status.
- 

## 30. Live Controller Monitoring (Center-Based)

Real-time operational states for assigned candidates. Summary counters:

- Total Registered.
- Duplicate Cancelled.
- Allocated / Unallocated.
- Not Logged In.
- Waiting for Start.
- Verification Required.

- Present/Verified.
- Absent.
- In Progress.
- Paused.
- Flagged.
- Submitted.
- Completed.
- Terminated/Invalidated.
- Slots Closed.
- Result Generated.
- Result Sent.

Candidate rows show operational status, center, slot and actions.

---

## 31. Student Dashboard (Website/App)

After ScorePlus login, the Pratham/Test area shows exam cards based on purchased package entitlements.

Cards can show test name, package name, package type (Online/Center-Based), language, scheduled date/time/slot, city and center (Center-Based), admit-card link, Live Exam status (upcoming, live, attended, absent, invalidated), result status, Practice availability, official score/rank where available and actions.

An exam stays visible even if the candidate did not attend the Live Exam, provided it belongs to a purchased package.

Center-Based cards never offer a Live attempt button on Website/App.

---

## 32. Result Calculation

Correct answers and marking rules come from the reused test/question setup. After attempt finalization the system calculates:

- Total questions.
- Attempted.
- Unattempted.
- Correct.
- Incorrect.
- Marks obtained.
- Maximum marks.
- Negative marks where existing rules apply.
- Final score.

Calculation must use the exact question/version/order assigned to that candidate.

---

# 33. Result and Answer-Analysis PDF

A detailed PDF is generated for each valid official attempt.

## 33.1 Summary

Candidate details, roll number, package and exam details, maximum marks, marks obtained, correct, incorrect, unattempted and other configured scoring information.

## 33.2 Question-Level Analysis

For each question: question, selected answer, correct answer, Correct/Incorrect/Unattempted status, marks awarded/deducted, and relevant images.

The purpose is to let the candidate see why a response got or lost marks.

---

# 34. Result Release

Do **not** create a new result-release scheduler. Reuse the existing configured release flow (e.g. immediate, scheduled, custom date/time as the existing system supports).

Result calculation and result release are separate: results can be calculated right after finalization while release follows configuration.

For a **Center-Based** individual exam, add one gate:

**All Assigned Slots Closed → Existing Result Release Configuration → Result Published**

- Absent candidates count as operationally closed and do not block release.
- Each included test has an independent lifecycle. The system does **not** wait for the whole package to finish before releasing an already completed individual exam.

For **Online**, the existing result release flow applies directly.

When released, the system can email the result/answer-analysis PDF. Email delivery status is tracked.

## 34.1 Slot Closure

A slot is **closed** when no candidate in it can still change their attempt.

- **Automatic close:** the system closes a slot once every assigned candidate is in a terminal state: Submitted, Auto Finalized, Absent, Terminated or Invalidated.
- **Stuck candidates:** candidates still in Disconnected or Recovery Required after the official end time (including any Emergency Pause extension) plus a configured grace period are **auto-finalized from server-side synced state**. This prevents results being blocked forever.

- **Force-close:** the Center Controller (only slots in their assigned exam/center/scope), Admin (slots within their authorized administrative scope) or Super Admin (any slot globally) can force-close a slot with a mandatory reason. Invigilators cannot. Any non-terminal candidates are auto-finalized from server state. Force-close is audited.
- Once closed, a slot cannot be reopened for new attempts.
- An exam's "All Assigned Slots Closed" gate is met when every slot of that exam, across all centers and dates, is closed.

## 34.2 Release Timing When the Gate Is Late

If the existing configured release time has already passed when the last slot closes (e.g. release set for 10 Oct 6 PM, last slot closes 10 Oct 8 PM), the result is **published immediately when the last slot closes**. The original delay is not recalculated. If the configured time is still in the future, the existing schedule applies unchanged.

## 34.3 Result Lock and Reprocessing

- Results have processing states.
- Finalized results are lockable.
- Reprocessing requires authorized action and a logged reason.
- Previous and revised processing references are auditable.

This prevents silent result changes.

---

# 35. Practice Mode

- Practice is always attempted on **Website/Mobile App**, for both Online and Center-Based packages.
  - Practice is **never** attempted on the Center-Based desktop app.
  - An exam becomes available for Practice after its official result is published/released for practice, according to the existing flow.
  - Practice uses the same questions, pattern, correct answers and scoring/negative-marking rules.
  - Practice does not use Live security overhead: no kiosk, camera proctoring, fraud escalation, Controller/invigilator monitoring or ranking.
  - Practice may show immediate scoring and answer analysis according to the existing flow.
  - Practice attempts never alter official Live results or ranking and must never overwrite official Live data.
  - Late purchasers and students absent from a Live Exam may receive eligible completed tests as Practice.
- 

# 36. Leaderboard

- Leaderboard/ranking uses **official Live Exam results only**.
- Practice attempts never create or change the official leaderboard.
- Invalidated/cancelled attempts are excluded.
- The leaderboard can show rank, approved display identity, official score and top performers.

- Students who took part see their official rank and relevant ranking information.
  - Students who did not take part (absent or late purchase) may view the published leaderboard but get no personal official rank.
  - If no ranking data exists, show: **Leaderboard not available yet.**
- 

# 37. Database Architecture and Data Isolation

## 37.1 MySQL

Existing Form Builder definition/configuration and unrelated existing legacy functionality remain in MySQL.

## 37.2 PostgreSQL

After CBT-enabled form submission, PostgreSQL is the **only** source of truth for Pratham/Test Series operational data for **both Online and Center-Based packages**, including:

- Form submissions/registrations and candidate profiles.
- Duplicate decisions.
- Payment/enrollment state and entitlements.
- Packages, package type and package/test mappings.
- Tests, language, question mappings and settings.
- City preferences.
- Cities, centers and capacity.
- Groups and slots.
- Allocations (tentative and finalized) and overrides.
- Conflicts.
- Finalization records.
- Roll numbers and credentials.
- Admit cards and delivery status.
- Controller/Invigilator accounts and assignments.
- Verification and attendance.
- Login/logout, session and device state.
- Attempts and responses.
- Randomization records.
- Queue/recovery state.
- Proctoring events, warnings, flags and controller decisions.
- Emergency pauses.
- Results, result items and generated documents.
- Email delivery status.
- Practice eligibility.
- Leaderboards.

- Audit logs.

There is **no duplicate submission copy in MySQL** and **no competing operational source of truth** (e.g. separate Online vs Center-Based stores).

## 37.3 Laravel Connectivity

Laravel keeps using MySQL for the existing product while the Pratham/CBT module uses PostgreSQL as a separate connection. The UI may be one Admin Panel, but the Pratham/CBT data source is PostgreSQL.

---

# 38. Audit Logging

Audit logging is mandatory, append-oriented and protected from ordinary editing.

Events include:

- Registration submitted.
- Duplicate detected/cancelled.
- Payment state changes, webhook receipt, reconciliation results, duplicate-payment flags, enrollment and entitlement.
- City preference submitted/changed.
- Allocation, reallocation and manual override (including Second Preference moves and date tweaks).
- Conflict detected/resolved.
- Allocation finalization.
- Rejected allocation due to full capacity.
- Credential generation.
- Admit-card generation and delivery.
- Emergency Reallocation, admit-card revocation and new version issue, student notification.
- Question delivery strategy selection and lock.
- Verification (Present/Verified), manual verification and Absent.
- Login/logout, device/session binding and device replacement.
- Waiting/start and Controller-authorized start.
- Answer sync milestones where useful.
- Connectivity loss/recovery; crash/reconnect.
- Proctoring warnings, flags and Controller flag decisions.
- Security-setting change.
- Emergency pause/resume.
- Submission and auto-finalization.
- Termination and post-submission invalidation.
- Slot closure (automatic, stuck-candidate auto-finalization, force-close).
- Result calculation, reprocessing, PDF generation, release and delivery.
- Practice release.

Each record captures, where relevant: event ID, actor (Admin/Controller/Invigilator/system), timestamp, event type, old/new state, reason, candidate/roll number, package ID, test/exam ID, device/session reference and supporting metadata.

---

## 39. Admin Exam Summary Dashboard

Per exam/package summary:

- Registered, Valid, Duplicate Cancelled, Paid, Unpaid/Incomplete.
- Online / Center-Based.
- Allocated, Pending, Unallocated, Conflicts.
- Capacity by center/date/slot.
- Allocation Finalized.
- Admit Cards Generated / Sent.
- Not Logged In, Waiting, In Progress, Flagged, Submitted, Completed, Absent, Terminated/Invalidated.
- Slots Closed.
- Results Processed / Released.
- Result Emails Sent / Failed.

Filters: package, exam, package type, city, center, date/slot, group, Controller, candidate status, flag status.

---

## 40. Notifications and Email

Transactional notifications for:

- Registration/payment confirmation.
- Allocation/admit card.
- Credential communication (Center-Based).
- Result/answer-analysis release.
- Important exam updates where configured.

Email failures must not disappear silently. Delivery status and retry/error info must be visible to authorized staff.

---

## 41. Security Requirements

- TLS for all client-server communication.
- Secure password hashing; no plaintext candidate passwords in PostgreSQL.
- Role-based access control; Controller/Invigilator scope limits.
- Server-authoritative exam timing.
- Session/token expiry and concurrent-session control.

- Server-side enforcement that Center-Based Live attempts are only accepted from the desktop app, never from Website/App.
- Secure local temporary exam storage and cache cleanup after safe finalization.
- API authentication, rate limiting where appropriate, input validation.
- Protection against ordinary API replay/tampering.
- Secure payment handling.
- Audit logging for privileged actions.
- Database backup strategy.
- Secrets outside source code.
- Production logging without passwords/tokens.

For high-stakes deployment, a formal security review and penetration test should be done before production.

---

## 42. Privacy Requirements

Where camera-based proctoring is enabled (Center-Based only):

- Clearly inform candidates.
- Request required permissions.
- Do not send continuous raw video when local detection is enough.
- Define retention for any retained snapshots/evidence.
- Limit and audit access to evidence.
- Follow applicable legal/organizational privacy rules.

Form data shown to invigilators is limited to what verification needs.

---

## 43. Reliability and Backup

PostgreSQL is operationally critical. Production needs:

- Automated backups.
- Point-in-time recovery where supported.
- Monitoring and database health alerts.
- Queue/state recovery.
- Defined retention.
- Disaster-recovery procedure.

The system must not rely on a candidate's machine as the only copy of exam progress.

---

## 44. Key State Models

## 44.1 Package Registration

Form Submitted → Duplicate Checked → Preferences Submitted (Center-Based) → Payment Initiated → Payment Pending → Payment Success → Entitled

**Exceptional:** Duplicate Cancelled, Payment Failed, Debited - Confirmation Pending (resolved by webhook or reconciliation), Incomplete/Unpaid (never allocated, never gets admit card).

## 44.2 Center-Based Exam Allocation

Unallocated → Tentatively Allocated → Conflict (warning) → Resolved → Finalized → Admit Card Generated (Locked, per exam)

**Exceptional:** Emergency Reallocated → Admit Card Revoked → New Admit Card Version Generated (Locked) (Super Admin only).

### 44.2A Slot

Scheduled → In Progress → Closed (automatic or force-close)

## 44.3 Center-Based Live Attempt

Admit Card Generated → Waiting → Present/Verified → Ready → In Progress → Submitted/Auto Finalized → Result Processing → Result Ready → Slots Closed → Result Released

**Exceptional:** Absent, Manual Verification Required, Flagged, Paused, Disconnected, Recovery Required, Terminated, Invalidated After Submission.

## 44.4 Online Live Attempt

Entitled → Scheduled → Live → Submitted/Auto Finalized → Result Released (existing flow)

**Exceptional:** Not Attempted (treated like absent — no official rank).

## 44.5 Practice

Result Released → Practice Available → Practice Attempt → Practice Score (never official).

All transitions are controlled and audit logged.

---

# 45. End-to-End Flows

## 45.1 Center-Based

**Account → CBT Form → Submit → PostgreSQL → Package-Scoped Duplicate Check → City Preferences → Payment → Entitlement → City/Center/Group/Slot Planning → Clash Check → Admin Adjustments → Finalize Allocation → Individual Admit Card → Assignment Lock → Center Attendance → Fresh Invigilator Verification → Existing Node.js CBT App → Secure Live Exam → Randomized Paper → Sync/Recovery → All Slots Closed → Existing Result Release Schedule → Official Result/Leaderboard → Practice on Website/App**

## 45.2 Online

**Account → CBT Form → Submit → PostgreSQL → Package-Scoped Duplicate Check → Payment → Entitlement → Scheduled Online Live Test → Website/Mobile App → Existing Result Release Flow → Official Result/Leaderboard → Practice on Website/Mobile App**

## 45.3 Diagram

## Existing ScorePlus Form Builder (MySQL)

|  
| Admin checks "Use this form as a CBT Test Form"  
v

Student (ScorePlus account) selects package on Website Test Series page  
or App Pratham card, fills form once per package, submits

|  
+----- PRATHAM / CBT DATA BOUNDARY (PostgreSQL) -----+  
v

### Package-Scoped Duplicate Check

|  
+-----+  
| Online | Center-Based  
| v  
| City Preferences  
v |  
Payment <-----+

### Package Entitlement

|  
+-----+  
| Online | Center-Based  
v v

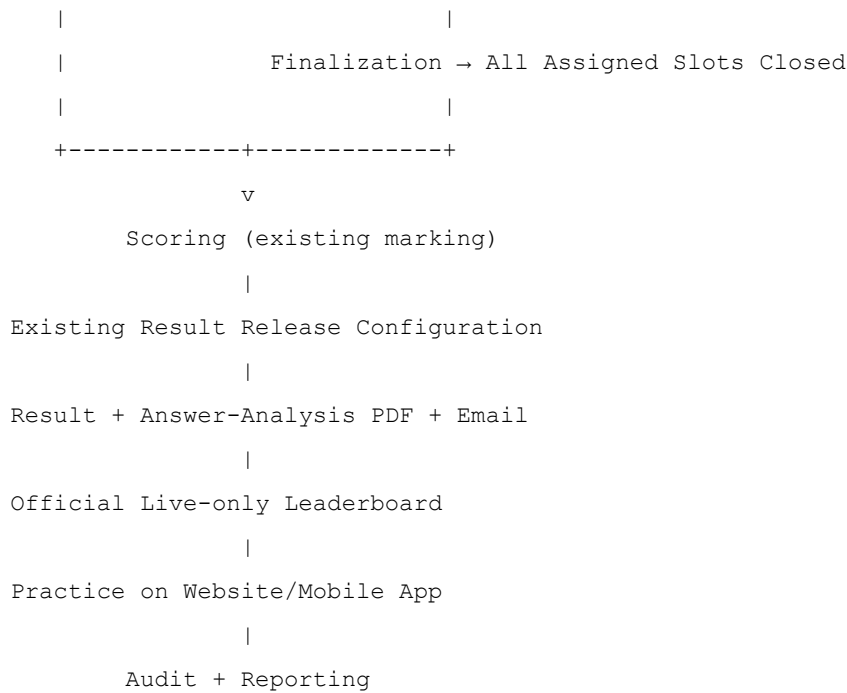
Scheduled Online City/Center/Group/Slot Planning

Live Test

(Website/App)

Clash Check (again after every manual change)

|  
| Admin Adjustments → Finalize Allocation  
|  
| Roll No + Password, Per-Exam Admit Card → LOCK  
|  
| Existing Admit-Card Delivery Schedule  
|  
| Center Attendance → Fresh Invigilator Verification  
| (or Absent)  
| Existing Node.js Windows CBT App  
|  
| Countdown → Kiosk → Randomized Paper  
|  
| Local Save + Server Queue Sync / Recovery  
| Optional Local Proctoring → Warning 1 → Warning 2  
| → Flag → Controller Review  
| Emergency Pause (+exact duration)



## 46. Suggested PostgreSQL Domain Model

Exact schema is an engineering decision. Logical entities similar to:

- pratham\_packages (with type: online / center\_based)
- pratham\_package\_tests
- pratham\_package\_cities
- cbt\_exams / cbt\_exam\_settings (incl. language)
- cbt\_questions / cbt\_exam\_questions
- cbt\_registrations
- cbt\_candidates
- cbt\_duplicate\_decisions
- cbt\_payments / cbt\_payment\_events (idempotent webhook log)
- cbt\_entitlements
- cbt\_city\_preferences
- cbt\_cities
- cbt\_centers (with capacity)
- cbt\_slots
- cbt\_groups
- cbt\_allocations (tentative/finalized/locked)
- cbt\_allocation\_conflicts
- cbt\_allocation\_finalizations
- cbt\_credentials
- cbt\_admit\_cards (versioned, with revoked status)

- cbt\_emergency\_reallocations
- cbt\_paper\_sets / cbt\_exam\_delivery\_strategy
- cbt\_controllers / cbt\_invigilators
- cbt\_staff\_assignments
- cbt\_verifications / cbt\_attendance
- cbt\_attempts
- cbt\_attempt\_question\_order
- cbt\_responses
- cbt\_response\_sync\_events
- cbt\_sessions / cbt\_devices
- cbt\_proctoring\_events / cbt\_warnings / cbt\_flags
- cbt\_controller\_actions
- cbt\_emergency\_pauses
- cbt\_slot\_closures
- cbt\_results / cbt\_result\_items
- cbt\_practice\_eligibility
- cbt\_leaderboards
- cbt\_generated\_documents
- cbt\_email\_deliveries
- cbt\_audit\_logs

Names are illustrative; engineering may normalize/rename them.

---

## 47. Suggested API Domains

Final endpoint design belongs to engineering. Broad domains:

### Student (Website/App)

- Catalogue and package details.
- Cart and payment (plus payment status check).
- Form submission, city preferences.
- Entitlements and dashboard.
- Allocation/admit-card/status view.
- Online Live Test: eligibility, countdown/time sync, payload, response sync, finalization.
- Practice attempt.
- Results, answer analysis, leaderboard.

### Center Desktop App

- Roll Number + Password authentication.
- Eligibility and verification status.

- Countdown/time sync.
- Exam payload.
- Response synchronization.
- Heartbeat/session status.
- Proctoring event submission.
- Pause state.
- Resume/recovery.
- Attempt finalization.
- Version check.

## Invigilator

- Assigned candidates per exam/slot.
- Form data for verification.
- Present/Verified or Absent.

## Center Controller

- Authentication and assigned scope.
- Candidate live statuses.
- Manual verification, start authorization.
- Flag review, continue/terminate/invalidate.
- Device-change approval.
- Proctoring configuration where authorized.
- Emergency pause/resume.
- Slot force-close (assigned scope only).

## Payment Gateway

- Idempotent webhook/callback receiver.
- Reconciliation (gateway status query).

## Admin

- Packages, tests, questions, language.
- Registrations, duplicates, payments, entitlements.
- Cities, centers, capacity, slots, groups.
- Allocation, conflicts, overrides, finalization.
- Emergency Reallocation (Super Admin only).
- Slot force-close (Admin: authorized scope; Super Admin: global).
- Question delivery strategy and paper sets.
- Payment status and manual re-check.
- Admit cards and credentials.
- Staff accounts and assignments.

- Result/practice configuration.
  - Leaderboards, reports, audit logs.
- 

## 48. UI/UX Guidelines

### Typography

**Primary Font: Poppins**

- Headings: Poppins SemiBold/Bold.
- UI labels: Poppins Medium.
- Body/question text: Poppins Regular/Medium.
- Timer and critical status: high-visibility Poppins SemiBold/Bold.

### Design Direction

- Professional, low-distraction exam interface.
  - High readability and strong contrast.
  - Clear state changes and consistent buttons.
  - NTA-inspired workflow with ScorePlus branding (no NTA branding).
  - Palette understandable through shape/status legend.
  - Security warnings noticeable but not chaotic.
  - Responsive to common desktop resolutions; edge-to-edge secure layout after activation (desktop).
  - Package cards clearly show Online vs Center-Based, language, schedule and cities.
  - Pratham card on App Home above Recent Videos / Top Videos.
- 

## 49. Functional Acceptance Criteria

Ready when, at minimum:

1. Pratham packages appear on the Website Test Series page and in the App Pratham card from one catalogue.
2. Admin can set each package to Online or Center-Based via dropdown; mixing is impossible.
3. Package cards show name, price, test count, subjects, schedule, language, type and (Center-Based) cities.
4. Students can buy multiple packages through a cart; individual tests cannot be bought.
5. Admin can mark an existing form as CBT using a checkbox; no new mandatory fields are forced.
6. Form is filled once per package.
7. Submission enters PostgreSQL with no duplicate copy in MySQL.
8. Duplicate detection is candidate + package scoped; different packages are never globally blocked.
9. Center-Based packages collect package-level city preferences.
10. Unpaid/incomplete registrations are never allocated or given admit cards.
11. Entitlement maps all included tests to the account.
12. Late purchase works: past exams → Practice where eligible, upcoming → Live, no retrospective rank.

13. Allocation keeps First Preference city using later dates/slots before Second Preference; Admin can move groups to Second Preference.
14. Physical center can differ per test inside the allocated city; capacity is respected.
15. Overlapping exams for the same student are prevented; clash check reruns after every manual reschedule.
16. Admin must Finalize Allocation before admit cards.
17. One admit card per Center-Based exam; date/slot/center locked after generation.
18. Center-Based candidates get dynamic roll number and password.
19. Invigilator verification is required for every Center-Based exam and uses only form data.
20. Absent candidates do not block slot closure or result release.
21. Candidate can log in early and see countdown with no question exposure; cannot begin before server-authorized start.
22. Controller can authorize an exceptional start without extra time.
23. Center-Based exam enters secure kiosk mode at start.
24. Online Live Tests run on Website/App with no kiosk, proctoring, invigilator or controller.
25. Center-Based Live attempts are impossible from Website/App.
26. Questions render text/images/options from the reused structure; language is fixed per test.
27. Question order is randomized reproducibly without per-slot papers.
28. Existing marking/negative-marking logic works.
29. NTA-inspired navigation/palette states work.
30. Responses survive temporary internet loss and app crash.
31. Same candidate cannot run concurrent active attempts.
32. Optional proctoring gives two warnings, then a Controller flag.
33. Controller can clear/continue/terminate and invalidate a submitted attempt with audited reason.
34. Emergency pause extends end time by the exact authorized duration, with audit trail.
35. Exam auto-finalizes by authoritative timing.
36. Result is calculated with configured answers and marking rules.
37. Detailed answer-analysis PDF is generated.
38. Center-Based result is released through the existing flow only after all assigned slots close; no wait for the whole package.
39. Practice runs only on Website/App and never changes official results/ranking.
40. Leaderboard uses official Live results only; non-participants see it without a personal rank.
41. Admin can view complete exam/package summary with capacity and allocation counts.
42. Critical events/actions appear in audit logs.
43. Pratham/CBT module reads/writes all operational data from PostgreSQL for both package types.
44. Home Mode through the desktop app does not exist.
45. Two simultaneous allocations to a slot with one seat left: exactly one succeeds; capacity is never exceeded.
46. Payment: duplicate webhooks grant entitlement once; a "debited but webhook pending" payment blocks a second payment and is resolved by reconciliation.
47. Moving a student to Second Preference moves all remaining not-yet-admit-carded exams of that package to the new city.
48. Admit-card lock is per exam: generating Test 1's card leaves Tests 2/3 editable.
49. Super Admin Emergency Reallocation revokes the old admit card, issues a new version, notifies the student and keeps full audit; normal Admins cannot do it.

50. Admin can select Shuffle Same Questions, Random Subset from Larger Pool or Multiple Paper Sets per exam; the choice is locked once the exam starts; each candidate's paper is reproducible.
  51. Slots auto-close when all candidates are terminal; stuck candidates auto-finalize after end time + grace; Center Controller (assigned slots only), Admin (authorized scope) and Super Admin (global) can force-close with reason and audit; Invigilator cannot, and a Controller cannot force-close a slot outside their scope.
  52. If the configured release time has passed when the last slot closes, the result publishes immediately at closure.
  53. An Absent candidate cannot be moved to another Live slot of the same exam.
  54. Center PC replacement: old session invalidated → new device authorized → answers restored → original end time preserved → full audit.
- 

# 50. Recommended Implementation Phases

## Phase 1 — Backend Foundation

- PostgreSQL connection and Pratham/CBT module in Laravel.
- Packages (type dropdown, language, cities), tests, entitlements.
- CBT form checkbox and submission handoff.
- Package-scoped deduplication.
- Audit foundation.

## Phase 2 — Catalogue, Purchase and Student Surfaces

- Website Test Series page package cards and cart.
- App Pratham card above Recent Videos / Top Videos.
- City preferences, payment, entitlement.
- Payment state machine, idempotent webhooks, reconciliation job.
- Student Dashboard Pratham area.
- Late-purchase handling.

## Phase 3 — Center-Based Allocation

- Cities, centers, capacity, slots, groups.
- Concurrency-safe allocation (no overbooking).
- Allocation with First-Preference priority and Second-Preference override.
- Super Admin Emergency Reallocation with admit-card versioning.
- Cross-package clash checks (allocation + every reschedule).
- Finalize Allocation.
- Credentials, per-exam admit cards, assignment lock, existing delivery schedule.

## Phase 4 — Online Live Tests

- Online Live Test on Website/App reusing existing Live → Result → Practice flow, backed by PostgreSQL.

## Phase 5 — Center-Based Examination Engine (existing desktop app)

- Remove Home Mode.
- Roll Number + Password login, countdown, server time.
- Kiosk mode, NTA-inspired UI, randomization.
- Question delivery strategies (shuffle / subset / paper sets).
- Local persistence, queue sync, recovery, auto-finalization.
- Session/device binding.

## Phase 6 — Center Operations

- Invigilator verification and attendance/absent.
- Controller live dashboard, authorized start, reconnect approval, device replacement.
- Slot closure (auto, stuck-candidate finalization, force-close).
- Flag actions, termination/invalidation.
- Emergency pause/resume with exact time extension.

## Phase 7 — Optional Proctoring (Center-Based)

- Local face presence, multiple faces, head/gaze logic, leaving frame, phone/object signal.
- Warning escalation and configuration.

## Phase 8 — Results, Practice and Leaderboard

- Scoring with existing marking.
- Slot-closure gate + existing result release (immediate publish if configured time already passed).
- Answer-analysis PDF and email.
- Result lock/reprocessing.
- Practice on Website/App.
- Live-only leaderboard.
- Summaries and reporting.

## Phase 9 — Hardening

- Center machine system check and app-version enforcement.
- Backup/recovery testing.
- Load testing (thousands of candidates across centers/slots).
- Failure simulation (power/network/crash).
- Security assessment and kiosk-escape testing.
- Privacy review.
- Production monitoring.

---

# 51. Locked Business Rules

1. Pratham and Test Series are the same ecosystem.
2. Individual tests are not sold separately.
3. One package is entirely Online or entirely Center-Based.
4. Package type is manually selected by Admin through a dropdown.
5. Both types use PostgreSQL operationally after CBT-form submission.
6. Duplicate detection is candidate + package scoped.
7. Form is completed once per package.
8. Center-Based city preference is package-level.
9. First Preference city has maximum priority; later same-city dates are preferred before Second Preference.
10. There is no maximum allowed delay for retaining First Preference.
11. Admin may manually move groups/students to Second Preference.
12. Exact physical center may differ across tests within the allocated city.
13. Manual rescheduling must rerun cross-package clash checks.
14. A 1–2 day gap between a student's Center-Based exams is preferred, not mandatory.
15. Allocation remains editable until admit-card generation.
16. Admin must finalize allocation before admit-card generation.
17. Admit card is per individual Center-Based exam.
18. After admit-card generation, date/slot/center cannot change, except through Super Admin-only Emergency Reallocation.
19. Verification is fresh for every individual Center-Based live exam.
20. Absent candidates do not block result release.
21. Authorized Emergency Pause time is added to the exam end time.
22. Center-Based Live = existing Node.js desktop app only.
23. Online Live = Website/Mobile App.
24. All Practice = Website/Mobile App.
25. Practice never changes official Live ranking/result.
26. Result release reuses existing scheduling logic after all slots of that individual exam close.
27. Language is predefined per test; multiple language versions use separate test entries.
28. Late purchase is allowed; past exams become Practice where eligible and upcoming exams remain Live.
29. Packages have no cancellation/refund flow in this scope.
30. A ScorePlus account is mandatory; existing login methods (Email/Password, Phone/OTP, Google, Truecaller) continue.
31. No separate Practice login page is created.
32. Students may purchase multiple packages through a cart-style flow.
33. Package purchase automatically maps all included tests to the student's account.
34. Leaderboard/ranking uses official Live Exam results only; a student who did not attend may view it but has no personal official rank.
35. Incomplete/unpaid registrations are never allocated and never get an admit card.
36. Center/date/slot capacity is enforced atomically and can never be exceeded, even by simultaneous Admin actions.

37. Entitlement is granted only on verified payment success; webhooks are idempotent; pending/debited payments are reconciled and never force a second payment.
  38. A move to Second Preference (or any city change) applies to all remaining not-yet-admit-carded Center-Based exams of the package.
  39. Admit-card lock is per exam, not package-wide.
  40. Emergency Reallocation after admit-card generation is a Super Admin-only exceptional action: mandatory reason, old admit card revoked, new assignment/admit-card version, student notified, complete audit history.
  41. Question delivery strategy is Admin-selected per exam (Shuffle Same Questions / Random Subset from Larger Pool / Multiple Paper Sets) and locked once the exam starts.
  42. Slots auto-close when all candidates are terminal; stuck candidates are auto-finalized from server state after end time + grace; Center Controller may force-close only assigned slots, Admin only slots within authorized Admin scope, Super Admin any slot globally — always with mandatory reason and full audit. Invigilators cannot force-close. Emergency Reallocation remains Super Admin-only.
  43. If the configured release time has already passed when the last slot closes, the result is published immediately at closure.
  44. An Absent candidate cannot be reassigned to another Live slot of the same exam; only Practice remains.
  45. Center PC replacement preserves answers and the original end time and invalidates the old device session.
- 

## 52. Critical Bug-Prevention Conditions

The implementation must specifically avoid these errors:

- Do not preserve the old Home Mode through the Windows desktop app.
- Do not apply kiosk/proctoring/invigilator restrictions to Online Live Tests.
- Do not allow Center-Based Live attempts from Website/Mobile App.
- Do not globally duplicate-block a student registered for different packages.
- Do not allocate unpaid/incomplete registrations.
- Do not change date/slot/center after admit-card generation, except through Super Admin Emergency Reallocation.
- Do not lock the whole package when one exam's admit card is generated.
- Do not let concurrent allocations exceed center/slot capacity.
- Do not grant entitlement twice for a repeated webhook, and do not ask a student to pay again while a payment is pending confirmation.
- Do not let a stuck Disconnected/Recovery Required candidate block slot closure forever.
- Do not reassign an Absent candidate to another Live slot of the same exam.
- Do not change an exam's question delivery strategy after it starts.
- Do not allow manual rescheduling to bypass clash validation.
- Do not wait for the full package to finish before releasing an individual completed test.
- Do not let absent candidates block release.
- Do not generate official rank from Practice attempts.
- Do not force verification documents absent from the Admin-configured form.
- Do not create a second Node.js CBT desktop application.
- Do not create separate competing data sources for Online vs Center-Based Pratham operational data.

- Do not confuse **package cancellation** with **individual exam-attempt invalidation/termination**. Package ownership cannot be cancelled in this scope, while an authorized center operator may still invalidate/terminate an individual attempt where the approved exam rules require it.

---

## 53. Final Architecture Rule

***MySQL remains responsible for the existing Form Builder configuration. Once the CBT-enabled form is submitted, PostgreSQL becomes the source of truth for Pratham/Test Series operational lifecycle data for both Online and Center-Based packages.***

***Center-Based Live Exams run only through the existing Node.js Windows CBT Desktop Application. Online Live Tests and every Practice Test run through the ScorePlus Website/Mobile App.***

***Pratham and Test Series are one product ecosystem.***

This boundary must stay consistent across backend, Laravel screens, APIs, Website, Mobile App, desktop app state, Controller/Invigilator operations, reporting and result processing.

---

## 54. Reconciliation / Final Check

This version intentionally replaces the old Home-vs-Center desktop architecture. The previous Home Mode through the Windows CBT app must not survive in implementation.

It also resolves the main bug-prone relationships found during review: package-scoped duplicate detection, city-level package allocation, physical-center flexibility, capacity/groups/slots, unlimited same-city delay, Admin Second-Preference override, cross-package conflict rechecking, explicit allocation finalization, post-admit locking, per-exam admit cards, per-exam invigilator verification, late purchase, absent handling, exact emergency-pause compensation, fixed per-test language, independent per-test result lifecycle, Practice isolation and Live-only leaderboard ranking.

The follow-up review also closed: concurrency-safe capacity, Super Admin Emergency Reallocation after admit-card generation, per-exam question delivery strategy for multi-slot leakage, slot-closure authority and stuck-candidate handling, late-gate release timing, payment lifecycle and reconciliation, package-wide effect of a city move, per-exam (not package-wide) admit-card lock, no Live reassignment after Absent, and center PC replacement.

No known functional contradiction from the reviewed requirements is intentionally left unresolved. Engineering details such as table names, API routes, queue technology and internal service boundaries may be chosen during implementation, but they must preserve these locked business rules.



